Describe the various instruments used in collecting data through Survey research.

Instruments Used in Collecting Data Through Survey Research

Survey research is one of the most widely used methods in social sciences, psychology, and management for collecting data about attitudes, opinions, behaviours, or facts from a defined population. To ensure accuracy and efficiency, various instruments are employed to gather reliable data. The main instruments are **sampling, questionnaires, and interviews**.

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1. Sampling

Sampling is often the first step in survey research because it determines who will provide the data. A **sample** is a subset of the population chosen to represent the entire group. Proper sampling ensures that findings can be generalised to the larger population.

- **Random sampling**: Every individual has an equal chance of being selected, reducing bias (e.g., lottery or fishbowl method).

- **Stratified sampling**: The population is divided into subgroups (e.g., departments in an organisation), and samples are drawn proportionately to ensure representation.

Example: If the goal is to assess job satisfaction in a company, selecting employees from all departments ensures balanced representation.

2. Questionnaires

Questionnaires are structured instruments consisting of written questions that respondents answer. They are efficient for collecting data from a large number of participants in less time.

Types of Questionnaire Administration:

- **Mail Survey**: Questionnaires are sent electronically or via post to respondents. It is inexpensive and time-saving but often suffers from low response rates.

- **Group-Administered Questionnaire**: Respondents gather at one place and fill out the questionnaire under the researcher's supervision. The advantage is that doubts can be clarified instantly.

- **Household Drop-off Survey**: The researcher personally delivers and later collects the questionnaire. This method increases response rates but requires more time and effort.

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3. Interviews

Interviews involve direct interaction between the researcher and respondent, making them more personal and flexible than questionnaires. They are particularly effective for collecting detailed or opinion-based data.

Types of Interviews:

- **Structured Interview**: Pre-planned questions are asked in a fixed order, ensuring consistency and comparability.

- **Unstructured Interview**: Conducted informally, where the respondent's answers guide the flow of questions, allowing deeper exploration of issues.

- **Telephonic Interview**: Conducted over the phone to save time and cost. However, it limits participation to individuals with phone access.

Conclusion

The choice of instrument in survey research depends on the research objectives, nature of the population, resources, and time constraints. **Sampling ensures representativeness, questionnaires allow large-scale data collection, and interviews provide depth and flexibility.** Often, researchers use a combination of these instruments to balance efficiency with depth and reliability.

8Q. Explain the different types of variables.

Types of Variables

In psychological and social science research, variables are central to describing, explaining, and predicting behaviour. A **variable** is any characteristic, attribute, or condition that can take different values across individuals, groups, or situations. Researchers classify variables in several ways depending on their role in a study.

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1. Stimulus, Organism, and Response Variables (S-O-R Model)

- **Stimulus Variables (S):** External factors or energy in the environment that affect behaviour, e.g., light, noise, or electric shock.

- **Organism Variables (O):** Internal characteristics of the individual such as age, anxiety, or heart rate.

- **Response Variables (R):** Observable behaviours or actions, e.g., pressing a lever, reaction time.
  Example: In an experiment where a rat jumps off a shock grid, the **shock** is the stimulus, the rat's **skin resistance** is the organismic variable, and the **latency to jump** is the response.

2. Independent and Dependent Variables

- **Independent Variable (IV):** The factor manipulated or selected to observe its effect (e.g., noise level, teaching method).

- **Dependent Variable (DV):** The outcome that changes in response to the IV (e.g., task performance, learning speed).

- **Types of IV:**

    - *Type E:* Directly manipulated (e.g., introducing continuous vs intermittent noise).

    - *Type S:* Manipulated by selection (e.g., age groups in productivity studies).

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3. Extraneous and Confounded Variables

- **Extraneous Variables:** Uncontrolled factors that may mask the IV–DV relationship. They can be:

    - *Organismic* (age, sex, intelligence),

    - *Situational* (temperature, noise),

    - *Sequential* (fatigue, practice effects).

- **Confounding Variables:** Variables that systematically vary with the IV, making causal inference difficult. For example, if adolescents form the experimental group and adults the control group in a TV-violence study, **age** becomes a confounding variable.

4. Active and Attribute Variables

- **Active Variables:** Manipulated by the researcher (e.g., reward, punishment, teaching methods).

- **Attribute Variables:** Pre-existing characteristics measured rather than manipulated (e.g., intelligence, gender, SES).

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5. Quantitative and Categorical Variables

- **Quantitative Variables:** Vary in amount, e.g., intelligence, speed of response.

- **Categorical Variables:** Vary in kind, e.g., gender, religion. These can be:

    - *Constant* (single value, e.g., water),

    - *Dichotomous* (two categories, e.g., yes/no),

    - *Polytomous* (multiple categories, e.g., religions).

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6. Continuous and Discrete Variables

- **Continuous Variables:** Measured with arbitrary precision, e.g., height, age, reaction time.

- **Discrete Variables:** Countable categories with gaps, e.g., family size, number of books.

Conclusion

Understanding variable types helps researchers design valid studies, control bias, and interpret findings correctly. Each type—whether **stimulus-response, independent-dependent, active-attribute, quantitative-categorical, or continuous-discrete**—plays a unique role in advancing psychological research.